

41. Chromosomal Alteration and the Development of Tumors. XXV

Quantitative Changes of Nucleolar Organizer Regions (NORs) in the Indian Spiny Mouse Tumors developed by 3-Methylcholanthrene^{*)}

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Indian spiny mouse, *Mus platythrix*, is characterized by 26 acrocentric chromosomes, and the size of the chromosome is generally larger than that of the house mouse. The nucleolar organizers of the Indian spiny mouse were found to be located near centromeres of three chromosome pairs, nos. 5, 8 and 12 (Yosida 1980). In the methylcholanthrene induced tumors (fibrosarcomas), Yosida (1981) reported the frequent occurrence of chromosome anomalies in the autosome pairs with the nucleolar organizer regions (NORs). In order to inquire into the problem as to whether the quantitative variation of the NORs occurs in tumor cells, and the high occurrence of abnormalities in the chromosomes with the NORs can observe in the other tumors, the karyotype analysis was undertaken in two 3-methylcholanthrene-induced tumors in the Indian spiny mouse. Another question whether a population shift occurs in deeply frozen tumor cells, or not, was also taken into account.

Material and methods. The Indian spiny mice, *Mus platythrix*, were collected in India in 1972, and then they have been bred in our laboratory for the experimental use. Tumor was induced by subcutaneous injection of 3-methylcholanthrene following the procedure described by Yosida (1981). Two tumors (MPMC-53 and MPMC-55) were developed and cultured *in vitro*. Tumor cells in early culture passages were used for this study. A batch of the MPMC-53 cultured cells was stored in the freezer at -80°C in the first culture passage by the routine procedure. Ten days after storage the cells were again cultured, and the karyotypes were analysed in comparison with those from continuous cultures. Urea treatment technique (Shiraishi and Yosida 1971) was applied for the G-band staining and the

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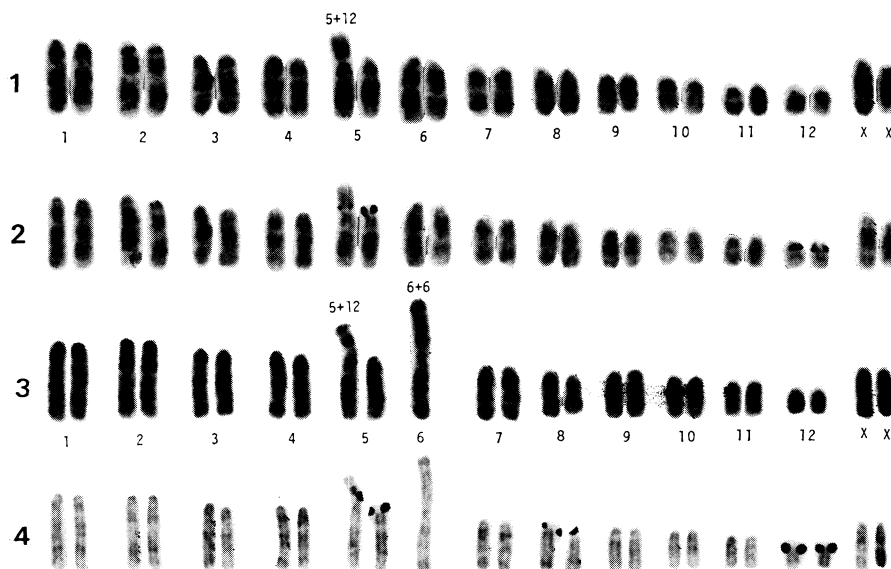
G-bands were removed by immersion of slides in 50% acetic acid, and then Ag-As staining method for the NORs (Bloom and Goodpasture 1976) was applied. Normal chromosomes were observed in cells from the tail-tip cultures in the tumor MPMC-55 bearing animal (Yosida *et al.* 1971).

Results. Normal somatic cells: Cells of the tail tip cultured from the animal bearing MPMC-55 showed 26 normal chromosomes, comprising 12 acrocentric autosome pairs (nos. 1 to 12) and acrocentric X and Y chromosomes. The Ag-As stained NORs in the normal cells were found to occur in chromosome nos. 5, 8 and 12 with considerably different frequencies, being 88.3% in pair no. 5, 14.2% in no. 8 and 47.2% in no. 12 (Table I).

MPMC-55: This tumor was characterized by a near-diploid stemline karyotype. In 22 karyotypes analysed, 16 cells (73%) had 26 chromosomes, leaving two cells with 25 chromosomes, three cells with 27 chromosomes and one cell with 28 chromosomes. Among them 12 cells showed translocations between nos. 5 and 12 (8 cells) (Figs. 1 and 2), between nos. 5 and 8 (one cell), and between some other chromosomes (3 cells). The frequencies of the NORs observed in 22 cells were 84.0% in no. 5 and 13.6% in no. 8, being nearly approximate to the normal value, while the frequency of the NORs in chromosome no. 12 was very high in tumor cells (109.1%). This outstandingly high frequency value of the NORs in no. 12 is attributable to trisomy of this chromosome (Table I).

Table I. Frequencies of NORs in tumors MPMC-55 and -53 in Indian spiny mouse

Tumors	Chrom. nos.	No. of NORs				Total no. of NORs (%)	No. of cells observed
		0	1	2	3		
Control Tail cells from animal bearing MPMC-55	5	0	22.6%	77.4%	0	88.8	53
	8	79.2%	13.2	7.5	0	14.2	53
	12	35.8	34.0	30.1	0	47.2	53
MPMC-55	5	0	31.8	68.2	0	84.0	22
	8	77.3	18.2	4.5	0	13.6	22
	12	9.0	9.0	36.4	45.5%	109.1	22
MPMC-53 Before freezing	5	0	33.3	60.0	6.6	86.6	15
	8	13.3	33.3	46.7	6.6	73.3	15
	12	13.3	20.0	60.0	6.6	80.0	15
MPMC-53 After freezing	5	0	4.0	92.0	4.0	100.0	25
	8	88.0	8.0	4.0	0	8.0	25
	12	4.0	12.0	76.0	8.0	94.0	25



Figs. 1-4. Karyotypes of methylcholanthrene-induced tumors in the Indian spiny mouse. 1: G-banding karyotype in an MPMC-55 tumor cell. Pair no. 12 is trisomic, and one of them is translocated to one of pair no. 5. 2: Ag-As staining of the above cell, showing NORs in nos. 5 and 12. 3: G-banding karyotype in an MPMC-53 tumor cell after frozen treatment. Translocations are observed between homologous chromosomes in no. 6 and between nos. 5 and 12. 4: Ag-As staining of the above cell, showing NORs in nos. 5, 8 and 12.

MPMC-53: In 26 cells studied, 15 cells (57.6%) were in a diploid range (24 to 27 chromosomes), and 10 cells (38.5%) in a tetraploid range (46 to 57 chromosomes) leaving one cell (3.8%) with 30 chromosomes. Among 15 cells of the diploid range, 10 cells showed translocations. The translocations occurred between no. 5 and no. 8 in 10 cells, and no. 8 and no. 12 in 2 cells. The latter translocation was present in duplication with the former one. The number of the NOR in nos. 5, 8 and 12 increased generally to 86.6, 76.6 and 80.0%, respectively (Table I).

Chromosomes observed in 68 cells in the frozen tumor showed that 57 cells (83.8%) were near-diploid (25 to 27 chromosomes), and 10 cells (14.7%) were near-tetraploid, leaving one (1.5%) with 30 chromosomes. The evidence seems to be attributable to a preferential selection for the diploid cells in the frozen tumor. The stemline karyotype of the frozen cells showed a marked alteration: 25 cells analysed had a new translocation between homologous chromosomes of no. 6 (Figs. 3 and 4). Translocation between nos. 5 and 8 which were frequent in the non-frozen cells was never observed in the cell population, but those between nos. 5 and 12 and between nos. 1 and

12 were found in two and three cells, respectively. Translocations between nos. 8 and 12, and nos. 5 and 5 were also counted in each one cell. The frequencies of NORs in nos. 5, 8 and 12 were 100.0, 8.0 and 90.0%, respectively. In the frozen cells the frequency of the organizers in nos. 5 and 12 was considerably higher, while it was lower in no. 8 (Table I).

Remarks. The banding karyotype analysis in the methylcholanthrene-induced tumors in the Indian spiny mouse revealed that chromosome anomalies of nos. 5, 8 and 12 with the NORs were considerably higher than the other chromosomes (Yosida 1981). In the tumors MPMC-55 and -53, the translocation including pair nos. 5, 8 and 12 was also very high in frequency. Important is to know whether or not the numerical change of the NORs occurs in tumor cells together with the increase of the translocation in the chromosome nos. 5, 8 and 12. The results of this study revealed, though the samples are limited, the number of the NORs clearly showed an increase in these tumors. Further investigations are requested for the final conclusion with regard to this matter.

In deeply frozen material for 10 days, the cell population of the MPMC-53 showed a marked change from that of the initial or *in vivo* passages: The frequency of diploid cells increased with the alteration of the karyotype. Yosida (1968) showed in cultured cells of the house mouse that the storage by freezing is more harmful to polyploid cells than to diploid cells. A similar situation was found in tumor cells of the spiny mouse: In the latter, the karyotype showed an alteration from that in the non-frozen specimens. This remains unknown at present.

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